OC Core 1.3.3 Reviewer Attack and Response Map

Bounded external-review scientific release

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Contents

1 Ontology of Continua Core 1.3.3

Version: 1.3.3 Tag: v1.3.3 DOI: 10.5281/zenodo.19965913 Zenodo record: https://zenodo.org/records/19965913

1.1 Release Boundary

OC Core 1.3.3 is a bounded external-review scientific release. It contains a typed model foundation, theorem/proof evidence, a Lean-checked subset, finite-model semantics, target-blind numeric reconstruction rows, comparator positioning, adversarial-review closure, and journal owner-review packets.

The release does not claim final completion of every future scientific projection. It does not submit journal packages. It does not claim universal superiority over all modern science. Those broader ambitions remain in the background research program and require additional evidence before public promotion.

The public GitHub and Zenodo publication is owner-approved for this release phase. Journal submissions, email campaigns, and Software Heritage actions require separate approval.

1.2 Adversarial Attack Matrix

This map lists concrete attack classes, artifact locations, severity, required repair, and closure evidence. A row is not considered closed merely because an artifact exists. - **row total:** 215 - **critical open total:** - **high open total:**

1.3 Prior-Art Comparator and Novelty Boundary

Comparator rows are positioning evidence, not a uniqueness proof for all possible theories. Each row states overlap, residual delta, and release-safe novelty boundary. - **register status:** - **row total:** 14

1.3.1 General System Theory

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{'oc_feature': 'cross-domain vocabulary', 'prior_art_overlap': 'YES', 'positioning_note': 'none; not claimed novel'}, {'oc_feature': 'typed proof/data/falsifier/owner-review release governance bundle', 'prior_art_overlap': 'NOT_OBSERVED_IN_ILLUSTRATIVE_SOURCE_NOT_ABSENCE_EVIDENCE', 'positioning_note': 'auditable release-governed scientific control plane'}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{'title': 'Ludwig von Bertalanffy, General System Theory', 'url': 'https://www.georgebraziller.com/general-systems-theory', 'source_date': '1968', 'inspected_on': '2026-04-28', 'local_protocol_snapshot_ref': 'comparators/source_snapshots/SRC-01-01.txt', 'local_protocol_snapshot_sha256': 'ac73ceab8486fb18e77fe2fd155ebf26 'search_query': ' "General System Theory" "organized wholes and cross-domain system language" release governance proof data falsifier', 'archive_status': 'LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROM

1.3.2 Autopoiesis

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{'oc_feature': 'self-producing living organization', 'prior_art_overlap': 'YES', 'positioning_note': 'none; not claimed novel'}, {'oc_feature': 'residue/rebirth/identity morphism separation with owner-review claim ledger', 'prior_art_overlap': 'NOT_OBSERVED_IN_ILLUSTRATIVE_SOURCE_NOT_ABSENC 'positioning_note': 'typed restart/identity equivocation blocker'}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{'title': 'Maturana and Varela, Autopoiesis and Cognition', 'url': 'https://link.springer.com/book/10.1007/ 94-009-8947-4', 'source_date': '1980', 'inspected_on': '2026-04-28', 'local_protocol_snapshot_ref': 'comparators/source_snapshots/SRC-02-01.txt', 'local_protocol_snapshot_sha256': '411971bd1277742eebee204a43f99c 'search_query': ' "Autopoiesis" "self-production and living organization" release governance proof data falsifier', 'archive_status': 'LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION'}]

1.3.3 Dynamical Systems

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.

- **feature tests:** [{‘oc_feature’: ‘state spaces, flows, iteration’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘smooth dynamics as one typed update specialization among proof/rewrite/hybrid updates’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘anti-universal-ODE typing rule’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘Encyclopedia of Mathematics, Dynamical system’, ‘url’: ‘https://encyclopediaofmath.org/wiki/Dy
‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots
03-01.txt’, ‘local_protocol_snapshot_sha256’: ‘3643c2233805d87a98f3b1148de0d461354dd8f06138f4a12c4476a3a9210665
‘search_query’: ‘ “Dynamical Systems” “state evolution and flows” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

1.3.4 Category and Topos Formalisms

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘typed objects, morphisms, internal logic’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘public-release theorem/evidence/falsifier lock over typed claims’, ‘prior_art_overlap’: ‘NOT_OBSERVED_IN_ILLUSTRATIVE_SOURCE_NOT_ABSENCE_EV
‘positioning_note’: ‘release-machine governance over scientific claim promotion’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘nLab, topos’, ‘url’: ‘https://ncatlab.org/nlab/show/topos’, ‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC-04-01.txt’, ‘local_protocol_snapshot_sha256’: ‘14f4b28a97898ae48909f877e08070d560998f850230daf171605c47e4d1f669’, ‘search_query’: ‘ “Category and Topos Formalisms” “typed objects, morphisms, categorical semantics” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_P

1.3.5 RAF Theory

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘autocatalytic closure and boundary relevance’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘K3 closure as adjacent K-level with demotion and release-proof witness rows’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘classifier-level irreducibility/demotion rule’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘Hordijk and Steel, Autocatalytic sets and boundaries’, ‘url’: ‘https://link.springer.com/article/10.1
014-0006-2’, ‘source_date’: ‘2015’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC-05-01.txt’, ‘local_protocol_snapshot_sha256’: ‘df9ee2dff8bcf86d318fb695a2e5a691
‘search_query’: ‘ “RAF Theory” “autocatalytic closure and boundaries” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

1.3.6 Complexity and Information Measures

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘information/complexity quantities’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘historical-axis versus effective-rank distinction inside OC K-level claims’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘typed anti-conflation theorem and finite witness’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘Stanford Encyclopedia of Philosophy, Information’, ‘url’: ‘https://plato.stanford.edu/entries/infor
‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots
06-01.txt’, ‘local_protocol_snapshot_sha256’: ‘004e1d650f029f27ba8cd0cf99cab8fcf07cb50e83bd7ab423e836b398544f25’, ‘search_query’: ‘ “Complexity and Information Measures” “information-theoretic and complexity quantities” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSEN

1.3.7 Causal and Identity Theories

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘persistence and identity criteria’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘residue/rebirth never promoted as identity without explicit invariant preservation’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘release claim-boundary lock’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘Stanford Encyclopedia of Philosophy, Identity Over Time’, ‘url’: ‘https://plato.stanford.edu/entrie time/’, ‘source_date’: ‘2026’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapsh 07-01.txt’, ‘local_protocol_snapshot_sha256’: ‘ffbde0b100aeb2986e68c25b2ef79e78a2a842c92d6f03f7b53969e4b6ccdef4’, ‘search_query’: ‘ “Causal and Identity Theories” “diachronic identity and persistence problems” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROM

1.3.8 Systems Engineering

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘requirements, verification, validation, lifecycle thinking’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘owner-gated owner-review scientific release state as theorem/evidence control plane’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘scientific publication lock integrated with claim ledger’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘INCOSE, Systems Engineering and System Definitions’, ‘url’: ‘https://www.incose.org/about-systems-engineering/system-and-se-definitions/’, ‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC-08-01.txt’, ‘local_protocol_snapshot_sha256’: ‘c279b39dbb5291d70c10d2493fbd7d5d2cd8cc1c1d067609b889ebd7373bf809’, ‘search_query’: ‘ “Systems Engineering” “verification, validation, lifecycle governance” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

1.3.9 Hybrid Systems

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘continuous/discrete hybrid transition systems’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘hybrid operator claim used to block universal differential overreach’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘claim-boundary role in OC operator theorem’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘Hybrid Systems III, Springer’, ‘url’: ‘https://link.springer.com/book/10.1007/BFb0031987’, ‘source_date’: ‘1996’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC 09-01.txt’, ‘local_protocol_snapshot_sha256’: ‘53321c329f66879221e61823275257e51d3c1e008cb5104bae6f7e9e2676a8b6’ ‘search_query’: ‘ “Hybrid Systems” “hybrid continuous/discrete transitions” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

1.3.10 Formal Methods and Lean

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{‘oc_feature’: ‘machine-checked proof development’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘Lean subset plus finite semantic witnesses plus release gates for public claim promotion’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘artifact-bound scientific release policy’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘Lean 4 official site’, ‘url’: ‘https://lean4.dev/’, ‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-28’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots/SRC-10-

01.txt', 'local_protocol_snapshot_sha256': '6d3b6cc5afbe2d8bae85a6cd7caf72f4f52d4a7195cc4d85fcd69346b8fb668b', 'search_query': ' "Formal Methods and Lean" "theorem proving and formal verification" release governance proof data falsifier', 'archive_status': 'LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTIO

1.3.11 Assurance Cases and Safety Cases

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{'oc_feature': 'structured argument that evidence supports a claim', 'prior_art_overlap': 'YES', 'positioning_note': 'none; not claimed novel'}, {'oc_feature': 'OC v12 claim ledger plus owner-review release gates', 'prior_art_overlap': 'PARTIAL', 'positioning_note': 'scientific-release control plane around theory claims'}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{'title': 'NASA System Safety Handbook, safety case/assurance case practice', 'url': 'https://www.nasa.gov/reference/system-safety-handbook/', 'source_date': 'reference', 'inspected_on': '2026-04-30', 'local_protocol_snapshot_ref': 'comparators/source_snapshots/SRC-11-01.txt', 'local_protocol_snapshot_sha256': '9c12be7e0cac872baf6e93547178bb857cee41b88d8c1d4e9001d548b0cba2b2', 'search_query': ' "Assurance Cases and Safety Cases" "assurance case argument/evidence structures" release governance proof data falsifier', 'archive_status': 'LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_P

1.3.12 Goal Structuring Notation / Argument Patterns

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{'oc_feature': 'claim -> argument -> evidence trace', 'prior_art_overlap': 'YES', 'positioning_note': 'none; not claimed novel'}, {'oc_feature': 'machine-generated attack matrix and finite-case route map', 'prior_art_overlap': 'PARTIAL', 'positioning_note': 'local scientific release binding; no uniqueness promoted'}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{'title': 'The Assurance Case Working Group and GSN community references', 'url': 'https://scsc.uk/scsc-141B', 'source_date': 'reference', 'inspected_on': '2026-04-30', 'local_protocol_snapshot_ref': 'comparators/source_snapshots/SRC-12-01.txt', 'local_protocol_snapshot_sha256': 'ee43f5a19310487c8c6ac885bfe9068ec6cb40e3e1c303c9edc54fce5913b861', 'search_query': ' "Goal Structuring Notation / Argument Patterns" "structured claim/evidence argument notation" release governance proof data falsifier', 'archive_status': 'LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION'}]

1.3.13 Requirements Traceability and V&V Matrices

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.
- **feature tests:** [{'oc_feature': 'requirement/claim to verification artifact traceability', 'prior_art_overlap': 'YES', 'positioning_note': 'none; not claimed novel'}, {'oc_feature': 'theorem-to-finite-case-to-Cerberus-finding closure ledger', 'prior_art_overlap': 'PARTIAL', 'positioning_note': 'local owner-review scientific release workflow'}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{'title': 'NASA Systems Engineering Handbook', 'url': 'https://www.nasa.gov/reference/nasa-systems-engineering-handbook/', 'source_date': 'reference', 'inspected_on': '2026-04-30', 'local_protocol_snapshot_ref': 'comparators/source_snapshots/SRC-13-01.txt', 'local_protocol_snapshot_sha256': '9c24dc9b859c4447b534b035086318467e6cc072cbc5d5b711b10073a9eab584', 'search_query': ' "Requirements Traceability and V&V Matrices" "requirements traceability and verification/validation matrices" release governance proof data falsifier', 'archive_status': 'LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROM

1.3.14 Scientific Workflow Provenance and Reproducibility Systems

- **absence test:** NOT_PROMOTED. Single-source absence is recorded only as a future search obligation, not as novelty evidence.

- **feature tests:** [{‘oc_feature’: ‘artifact metadata, provenance, and reproducibility package’, ‘prior_art_overlap’: ‘YES’, ‘positioning_note’: ‘none; not claimed novel’}, {‘oc_feature’: ‘strict owner-review theorem/replay/Cerberus gate stack’, ‘prior_art_overlap’: ‘PARTIAL’, ‘positioning_note’: ‘local policy coupling; no uniqueness promoted’}]
- **uniqueness claim status:** NOT_PROMOTED_PRIOR_ART_POSITIONING_ONLY
- **source refs:** [{‘title’: ‘RO-Crate specification’, ‘url’: ‘https://www.researchobject.org/ro-crate/’, ‘source_date’: ‘reference’, ‘inspected_on’: ‘2026-04-30’, ‘local_protocol_snapshot_ref’: ‘comparators/source_snapshots 14-01.txt’, ‘local_protocol_snapshot_sha256’: ‘b71da73a5a5235fca9d08148a048225ad33f79da1bfc73eaee920e3b9d0daff5’, ‘search_query’: ‘ “Scientific Workflow Provenance and Reproducibility Systems” “workflow provenance, reproducibility metadata, and research-object packaging” release governance proof data falsifier’, ‘archive_status’: ‘LOCAL_PROTOCOL_CAPSULE_ONLY_NO_ABSENCE_PROMOTION’}]

1.4 Phenomenon Coverage and Limits

The phenomenon matrix records model cards and evidence routes. Rows with illustrative or protocol-ready status are not promoted as complete phenomenon explanations. - **row total:** 15 - **state:**

1.4.1 P001

- **observable:** same-cell raw pair is not distinguished; cross-cell quotient pair is distinguished
- **negative control:** FM-T133-K0-RES-NEG
- **falsifier:** If a same-rho-cell raw pair is accepted as resolution-distinguished, the K0 resolution boundary fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.4.2 P002

- **observable:** death blocks live status; rebirth source is residue and target is a new live token
- **negative control:** FM-T133-OMEGA-STATUS-NEG
- **falsifier:** If death=true and live=true are accepted together, or rebirth targets the original identity token, the lifecycle theorem fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.4.3 P003

- **observable:** live label fails without cycle or maintenance support
- **negative control:** FM-T133-CYCLE-NEG
- **falsifier:** If live=true is accepted with cycle_mode=none and maintenance support unavailable, the liveness route fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.4.4 P004

- **observable:** classifier failure equals declared failure predicate; metric wording is allowed only when a measure is declared
- **negative control:** FM-T133-BOUNDARY-NEG
- **falsifier:** If metric boundary language is accepted without metric_measure_declared=true, the boundary specialization fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.4.5 P005

- **observable:** proof/rewrite update is accepted only as a typed non-smooth transition

- **negative control:** FM-T133-HYBRID-PROOF-UPDATE-NEG
- **falsifier:** If a proof/rewrite state can request a derivative without a smooth chart, the operator boundary fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.4.6 P006

- **observable:** historical activation remains while effective rank drops
- **negative control:** FM-T133-DIM-NEG
- **falsifier:** If a historical axis can decrease when effective rank drops, the dimension theorem fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.4.7 P007

- **observable:** k=0 is licensed by declared zero-cause, not empty state set
- **negative control:** FM-T133-K-ZERO-NEG
- **falsifier:** If k=0 is accepted with nonempty support and no declared zero-cause, the k-zero theorem fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.4.8 P008

- **observable:** closure cannot be reduced when production witness remains observable
- **negative control:** FM-KLEVEL-K2_to_K3-NEG
- **falsifier:** If K2->K3 reduction preserves the production witness while still demoting, the closure-like K transition fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.4.9 P009

- **observable:** K6->K7 transition fails reduction when role witness changes verdict
- **negative control:** FM-KLEVEL-K6_to_K7-NEG
- **falsifier:** If a role/norm witness changes allowed action but the K6->K7 reduction still passes, the institution model card fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.4.10 P010

- **observable:** K8->K9 transition fails reduction when claim revision is enabled
- **negative control:** FM-KLEVEL-K8_to_K9-NEG
- **falsifier:** If claim/evidence revision changes verdict but K8->K9 reduction still passes, the theory-change card fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.4.11 P011

- **observable:** self-application is accepted only through typed transition witness
- **negative control:** FM-KLEVEL-K9_to_K10-NEG
- **falsifier:** If self-application is admitted without the K9->K10 typed transition witness, the recursion card fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.4.12 P012

- **observable:** public action is rejected while global owner-review or any channel lock remains closed
- **negative control:** ADV-NOSEND-PUBLISH-HYPOTHETICAL-OWNER-APPROVED-CONTROL
- **falsifier:** If publish is allowed while owner approval is absent or any channel lock remains not authorized, the owner-review card fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.4.13 P013

- **observable:** reduction fails exactly when retained witness stays observable
- **negative control:** FM-T133-KLEVEL-NEG
- **falsifier:** If any adjacent K retained-witness row lacks its demotion control, the K-collapse card fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.4.14 P014

- **observable:** component removal changes declared verdict in semantic finite runner
- **negative control:** FM-T133-MIN-NEG
- **falsifier:** If a tuple component can be removed without the semantic finite verdict changing, the minimality card fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.4.15 P015

- **observable:** identity continuation is accepted only when endpoint-bound identity evidence is present
- **negative control:** FM-T133-ID-NEG
- **falsifier:** If residue or rebirth evidence is accepted as identity continuation without endpoint-bound identity evidence, the identity theorem fails.
- **claim boundary:** This is an internal release-consistency illustration and is excluded from phenomenon coverage totals until a domain model/evaluator is added.

1.5 Claim Governance

The public claim surface is intentionally bounded: a row is promoted only when it is evidence-bound, review-clean, and explicitly scoped. Universal closure or all-domain superiority language is not promoted by this release. - **claim total:** 18 - **unsupported promoted total:** 0 - **scientific promotion allowed total:** 10 - **absolute overclaim policy:** BLOCK_PUBLIC_PROMOTION

1.5.1 T133-K0-RES

- **claim:** K0 support is treated as a owner-review formal release-consistency check over declared resolution quotients: same-resolution states are not distinguished, and a finite countermodel shows raw separation need not induce resolution distinction. It is not promoted as an independent novelty or scientific theorem in v12.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-K0-RES.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.5.2 T133-OMEGA-STATUS

- **claim:** Death blocks live status; residue and rebirth are token-bound evidence relations with distinct class-specific endpoint rules: residue separates source from residue while returning to the source endpoint,

and rebirth separates source, residue, and new target tokens. Rebirth is non-identity unless endpoint-bound identity evidence has identity class, declared invariant preservation, no residue token, and equal source/target endpoint evidence. No categorical Hom/composition theorem is promoted.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-OMEGA-STATUS.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.5.3 T133-K-ZERO

- **claim:** Continuumness zero requires live support, an independently clear obstruction ledger, and a declared zero-cause family; a zero-cause label alone does not compute k=0.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-K-ZERO.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.5.4 T133-BOUNDARY

- **claim:** Metric thresholds are a specialization of typed classifier boundaries.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-BOUNDARY.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.5.5 T133-HYBRID

- **claim:** OC operators are typed update semantics; chart-labelled flow-one notation is admitted only for declared chart records, while proof/rewrite and guard/reset updates remain first-class non-smooth cases. No differentiability or ODE-solution theorem is promoted in v12.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-HYBRID.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.5.6 T133-DIM

- **claim:** Historical axis activation and effective working rank are kept as distinct owner-review formal release-consistency fields; a finite/Lean witness shows compatibility of monotone historical bookkeeping with decreasing effective rank, but v12 does not promote an independent scientific dimension theorem.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-DIM.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.5.7 T133-CYCLE

- **claim:** Declared eligible-live status requires an explicit cycle mode or non-vacuous maintenance predicate.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-CYCLE.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.5.8 T133-ID

- **claim:** Identity continuation requires endpoint-bound identity evidence: identity class, declared invariant preservation, no residue token, equal source/target endpoint evidence, lifecycle identity-invariant truth,

and typed source/target binding. Residue and rebirth evidence classes do not become identity continuation merely by preserving some invariants. No categorical Hom/composition theorem is promoted.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-ID.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.5.9 T133-MIN

- **claim:** Within the declared v12 release tuple semantics, each tuple component has a one-field semantic keep/drop witness that changes the release verdict.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-MIN.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.5.10 T133-KLEVEL

- **claim:** Every declared adjacent K-level transition K0->K12 has a release-atlas row, retained-witness evaluator check, executable finite row, and inert-witness demotion control inside the v12 release classifier; independent semantic irreducibility beyond this declared classifier is a future proof obligation, not a promoted v12 theorem.
- **support:** LEAN_SUBSET_STRUCTURED_PROOF_FINITE_WITNESS
- **evidence:** proofs/proof_sheets/T133-KLEVEL.md
- **status:** PROMOTED_BOUNDED_PUBLIC_RELEASE_V12
- **scope limit:** Bounded to stated theorem assumptions, finite witnesses, and public falsifier boundary.

1.5.11 OC133-NUM-PHYS-C

- **claim:** calibration replay of an official constant, not a new law of physics
- **support:** NUMERIC_REPLAY_QA_WITH_BASELINE_NEGATIVE_CONTROL_FALSIFIER
- **evidence:** validation/numeric_replay_qa/OC133_NUMERIC_REPLAY_QA_TABLE.json
- **status:** QUARANTINED_REPLAY_QA_NOT_PROMOTED_V12
- **scope limit:** This is replay QA and falsifier plumbing, not empirical theory promotion.

1.5.12 OC133-NUM-CHEM-WEBBOOK-H2O

- **claim:** NIST Chemistry WebBook molecular-weight field replay for water only
- **support:** NUMERIC_REPLAY_QA_WITH_BASELINE_NEGATIVE_CONTROL_FALSIFIER
- **evidence:** validation/numeric_replay_qa/OC133_NUMERIC_REPLAY_QA_TABLE.json
- **status:** QUARANTINED_REPLAY_QA_NOT_PROMOTED_V12
- **scope limit:** This is replay QA and falsifier plumbing, not empirical theory promotion.

1.5.13 OC133-NUM-CHEM-H2O

- **claim:** PubChem molecular-weight field replay for water only
- **support:** NUMERIC_REPLAY_QA_WITH_BASELINE_NEGATIVE_CONTROL_FALSIFIER
- **evidence:** validation/numeric_replay_qa/OC133_NUMERIC_REPLAY_QA_TABLE.json
- **status:** QUARANTINED_REPLAY_QA_NOT_PROMOTED_V12
- **scope limit:** This is replay QA and falsifier plumbing, not empirical theory promotion.

1.5.14 OC133-NUM-BIO-GEO-COUNT

- **claim:** official GEO query-count replay; no organism-wide mechanism claim
- **support:** NUMERIC_REPLAY_QA_WITH_BASELINE_NEGATIVE_CONTROL_FALSIFIER
- **evidence:** validation/numeric_replay_qa/OC133_NUMERIC_REPLAY_QA_TABLE.json
- **status:** QUARANTINED_REPLAY_QA_NOT_PROMOTED_V12

- **scope limit:** This is replay QA and falsifier plumbing, not empirical theory promotion.

1.5.15 OC133-NUM-SYS-WDI-GDP

- **claim:** retrospective WDI GDP snapshot replay QA with descriptive comparator, no prediction or superiority claim
- **support:** NUMERIC_REPLAY_QA_WITH_BASELINE_NEGATIVE_CONTROL_FALSIFIER
- **evidence:** validation/numeric_replay_qa/OC133_NUMERIC_REPLAY_QA_TABLE.json
- **status:** QUARANTINED_REPLAY_QA_NOT_PROMOTED_V12
- **scope limit:** This is replay QA and falsifier plumbing, not empirical theory promotion.

1.5.16 OC133-NUM-MATH-FINITE

- **claim:** finite witness acceptance count for machine-checked v12 theorem inventory rows; never public-promotion, empirical, or prediction support in v12 unless a future prospective protocol explicitly flips release_promotion_allowed, prediction_support_allowed, and empirical_support_allowed after review
- **support:** NUMERIC_REPLAY_QA_WITH_BASELINE_NEGATIVE_CONTROL_FALSIFIER
- **evidence:** validation/numeric_replay_qa/OC133_NUMERIC_REPLAY_QA_TABLE.json
- **status:** QUARANTINED_REPLAY_QA_NOT_PROMOTED_V12
- **scope limit:** This is replay QA and falsifier plumbing, not empirical theory promotion.

1.5.17 OC133-NOVELTY-001

- **claim:** Prior-art comparison is an illustrative bounded positioning note; uniqueness, priority, and absence are not promoted until a systematic search exists.
- **support:** ILLUSTRATIVE_PRIOR_ART_POSITIONING_ONLY_NO_UNIQUENESS_PROMOTION
- **evidence:** comparators/OC_1_3_3_NOVELTY_AND_PRIORITY_REGISTER.json
- **status:** NOT_PROMOTED_RESEARCH_NOTE_V12
- **scope limit:** No uniqueness, priority, absence, or invention claim is promoted by this row; systematic search remains future work.

1.5.18 OC133-NOSEND-001

- **claim:** Publication, DOI minting, repository release, deposit, and journal submission remain locked until both separate owner approval and an explicit manifest/channel unlock are present.
- **support:** OWNER_GATED_OWNER_REVIEW_LOCKED_CONTROL_PLANE
- **evidence:** releases/oc_core_1_3_3/editorial/OC_CORE_1_3_3_PUBLISH_MANIFEST_DRAFT.json
- **status:** GOVERNANCE_CONTROL_OWNER_REVIEW_LOCKED_NOT_SCIENTIFIC_PROMOTION
- **scope limit:** Local owner-review package only; this is not a real-world enforcement guarantee against manual or external publication. Owner approval alone is insufficient unless deposit-ready metadata, a public-record target, owner-review unlock, and every channel lock are all separately true.

1.6 Theorem and Proof Registry

The release promotes bounded theorem claims only where the theorem registry binds a proof sheet, Lean reference, finite witness route, and explicit scope boundary. Broad full-science and universal superiority claims stay outside the promoted release surface. - **registered theorem total:** 10 - **machine-checked subset total:** 10 - **adversarial blocker total:** 0

1.6.1 1. T133-K0-RES: K0 resolution-relative distinguishability theorem

- **claim boundary:** K0 support is treated as a owner-review formal release-consistency check over declared resolution quotients: same-resolution states are not distinguished, and a finite countermodel shows raw separation need not induce resolution distinction. It is not promoted as an independent novelty or scientific theorem in v12.
- **evidence ref:** appendix/OC_1_3_3_K0_RESOLUTION_FOUNDATION.tex
- **proof sheet:** proofs/proof_sheets/T133-K0-RES.md

- **Lean ref:** formal/lean/OC133V12.lean::k0_countermodel_raw_separation_not_resolution_distinction
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** A proof that assumes every pair of raw real states is epsilon-separated is outside v12 and fails G33.

1.6.2 2. T133-OMEGA-STATUS: Typed liveness, death, residue, and rebirth evidence-consistency theorem

- **claim boundary:** Death blocks live status; residue and rebirth are token-bound evidence relations with distinct class-specific endpoint rules: residue separates source from residue while returning to the source endpoint, and rebirth separates source, residue, and new target tokens. Rebirth is non-identity unless endpoint-bound identity evidence has identity class, declared invariant preservation, no residue token, and equal source/target endpoint evidence. No categorical Hom/composition theorem is promoted.
- **evidence ref:** content/OC_1_3_3_TYPED_FOUNDATION.tex
- **proof sheet:** proofs/proof_sheets/T133-OMEGA-STATUS.md
- **Lean ref:** formal/lean/OC133V12.lean::lifecycle_residue_rebirth_morphism_boundary
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** Any claim reading residue-preserving restart as same-identity survival is rejected unless endpoint-bound identity evidence is supplied.

1.6.3 3. T133-K-ZERO: Continuumness zero obstruction theorem

- **claim boundary:** Continuumness zero requires live support, an independently clear obstruction ledger, and a declared zero-cause family; a zero-cause label alone does not compute k=0.
- **evidence ref:** appendix/OC_1_3_3_CONTINUUMNESS_FUNCTIONALS.tex
- **proof sheet:** proofs/proof_sheets/T133-K-ZERO.md
- **Lean ref:** formal/lean/OC133V12.lean::continuumness_zero_case_iff_declared_zero_cause_with_support
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** A realization with undeclared zero-cause, missing live support, or any active obstruction may not promote k=0.

1.6.4 4. T133-BOUNDARY: Metric-threshold boundary specialization theorem

- **claim boundary:** Metric thresholds are a specialization of typed classifier boundaries.
- **evidence ref:** appendix/OC_1_3_3_BOUNDARY_REPRESENTATION_THEOREM.tex
- **proof sheet:** proofs/proof_sheets/T133-BOUNDARY.md
- **Lean ref:** formal/lean/OC133V12.lean::metric_boundary_specialization
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** A social or logical boundary represented numerically without a measurement rule is blocked by G40.

1.6.5 5. T133-HYBRID: Typed update and chart-labelled operator semantics theorem

- **claim boundary:** OC operators are typed update semantics; chart-labelled flow-one notation is admitted only for declared chart records, while proof/rewrite and guard/reset updates remain first-class non-smooth cases. No differentiability or ODE-solution theorem is promoted in v12.
- **evidence ref:** content/OC_1_3_3_OPERATOR_SEMANTICS.tex
- **proof sheet:** proofs/proof_sheets/T133-HYBRID.md
- **Lean ref:** formal/lean/OC133V12.lean::integrated_operator_semantics
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** Any route treating a chart token as differentiability, manifold, vector-field, or ODE-solution evidence is outside the v12 operator theorem and remains a future lint/proof obligation.

1.6.6 6. T133-DIM: Historical axis and effective-rank compatibility theorem

- **claim boundary:** Historical axis activation and effective working rank are kept as distinct owner-review formal release-consistency fields; a finite/Lean witness shows compatibility of monotone historical bookkeeping with decreasing effective rank, but v12 does not promote an independent scientific dimension theorem.

- **evidence ref:** appendix/OC_1_3_3_K_LEVEL_IRREDUCIBILITY_ATLAS.tex
- **proof sheet:** proofs/proof_sheets/T133-DIM.md
- **Lean ref:** formal/lean/OC133V12.lean::historical_axis_survives_rank_drop
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** A K-level is demotable only when the alleged new axis has no witness and no observable consequence.

1.6.7 7. T133-CYCLE: Live-status cycle-mode requirement theorem

- **claim boundary:** Declared eligible-live status requires an explicit cycle mode or non-vacuous maintenance predicate.
- **evidence ref:** content/OC_1_3_3_CYCLE_TAXONOMY.tex
- **proof sheet:** proofs/proof_sheets/T133-CYCLE.md
- **Lean ref:** formal/lean/OC133V12.lean::eligible_live_requires_cycle_or_maintenance
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** An artifact that never updates, replays, checks, or maintains itself is archive residue, not live continuum.

1.6.8 8. T133-ID: Identity, residue, and rebirth evidence-classification theorem

- **claim boundary:** Identity continuation requires endpoint-bound identity evidence: identity class, declared invariant preservation, no residue token, equal source/target endpoint evidence, lifecycle identity-invariant truth, and typed source/target binding. Residue and rebirth evidence classes do not become identity continuation merely by preserving some invariants. No categorical Hom/composition theorem is promoted.
- **evidence ref:** appendix/OC_1_3_3_IDENTITY_RESIDUE_REBIRTH_CLASSIFICATION.tex
- **proof sheet:** proofs/proof_sheets/T133-ID.md
- **Lean ref:** formal/lean/OC133V12.lean::endpoint_bound_identity_classification
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** Any public claim reading rebirth as literal same-identity survival is blocked.

1.6.9 9. T133-MIN: Declared semantic-verdict component independence theorem

- **claim boundary:** Within the declared v12 release tuple semantics, each tuple component has a one-field semantic keep/drop witness that changes the release verdict.
- **evidence ref:** appendix/OC_1_3_3_GLOBAL_MINIMALITY_WITNESSES.tex
- **proof sheet:** proofs/proof_sheets/T133-MIN.md
- **Lean ref:** formal/lean/OC133V12.lean::release_tuple_semantic_component_irredundant
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** If any declared tuple component can be removed while `FM-MIN-*` still returns PASS for the declared semantic verdict suite, the minimality card fails.

1.6.10 10. T133-KLEVEL: Declared adjacent K-level atlas/evaluator consistency theorem

- **claim boundary:** Every declared adjacent K-level transition K0->K12 has a release-atlas row, retained-witness evaluator check, executable finite row, and inert-witness demotion control inside the v12 release classifier; independent semantic irreducibility beyond this declared classifier is a future proof obligation, not a promoted v12 theorem.
- **evidence ref:** appendix/OC_1_3_3_K_LEVEL_IRREDUCIBILITY_ATLAS.tex
- **proof sheet:** proofs/proof_sheets/T133-KLEVEL.md
- **Lean ref:** formal/lean/OC133V12.lean::release_atlas_manifest_has_total_finite_case_coverage
- **evidence ceiling:** BOUNDED_SCIENTIFIC_THEOREM_WITH_LEAN_SUBSET_AND_FINITE_WITNESS
- **scope limit:** If any adjacent K row lacks row identity, retained-witness evaluator failure, or inert-witness demotion control, the finite negative control fails; independent semantic irreducibility remains unpromoted unless separately proven.

1.7 Lean Formalization Subset

The Lean subset is a machine-checked subset of the OC 1.3.3 theorem surface. The release does not claim that every mathematical or empirical statement is fully formalized in Lean. **- certificate state: - theorem ref total:** 10 **- missing theorem ref total: - Lean file:** formal/lean/OC133V12.lean

Selected declaration excerpt:

```
namespace OC133V12

inductive Status where
 | pass
 | fail
deriving DecidableEq, Repr

inductive CycleMode where
 | maintenance
 | renewal
 | replay
 | regulatory
 | degenerate
deriving DecidableEq, Repr

inductive MorphismClass where
 | identity
 | residue
 | rebirth
deriving DecidableEq, Repr

inductive Component where
 | carrier
 | realization
 | lawfulPossibility
 | liveness
 | residue
 | morphisms
 | boundaries
 | operators
 | cycles
 | dimension
 | kFunctional
deriving DecidableEq, Repr

inductive AdjacentK where
 | k0_k1
 | k1_k2
 | k2_k3
 | k3_k4
 | k4_k5
 | k5_k6
 | k6_k7
 | k7_k8
 | k8_k9
 | k9_k10
 | k10_k11
 | k11_k12
deriving DecidableEq, Repr
```

```
def boolStatus (b : Bool) : Status :=
 if b then Status.pass else Status.fail

structure Resolution (S : Type) where
 cell : S -> Nat

structure RawSeparation (S : Type) where
 separated : S -> S -> Bool

def sameCell {S : Type} (rho : Resolution S) (a b : S) : Prop :=
 rho.cell a = rho.cell b

def distinguished {S : Type} (rho : Resolution S) (a b : S) : Prop :=
 rho.cell a != rho.cell b

theorem k0_same_cell_not_distinguished {S : Type} (rho : Resolution S) (a b : S) :
 sameCell rho a b -> distinguished rho a b = False := by
 intro h
 unfold distinguished
 rw [h]
 simp

theorem k0_distinguished_requires_resolved_delta {S : Type} (rho : Resolution S) (a b : S) :
 distinguished rho a b -> sameCell rho a b -> False := by
 intro hd hs
 rw [k0_same_cell_not_distinguished rho a b hs] at hd
 exact hd

theorem k0_resolution_does_not_force_raw_separation {S : Type}
 (rho : Resolution S) (raw : RawSeparation S) (a b : S) :
 sameCell rho a b -> raw.separated a b = true -> distinguished rho a b = False := by
 intro hs _
 exact k0_same_cell_not_distinguished rho a b hs

inductive K0CounterPoint where
 | a
 | b
deriving DecidableEq, Repr

def k0CounterResolution : Resolution K0CounterPoint :=
 { cell := fun _ => 0 }

def k0CounterRawSeparation : RawSeparation K0CounterPoint :=
 { separated := fun x y => decide (x != y) }

theorem k0_countermodel_raw_separation_not_resolution_distinction :
 sameCell k0CounterResolution K0CounterPoint.a K0CounterPoint.b /\
 k0CounterRawSeparation.separated K0CounterPoint.a K0CounterPoint.b = true /\
 distinguished k0CounterResolution K0CounterPoint.a K0CounterPoint.b = False := by
 exact And.intro rfl (And.intro (by decide) (k0_same_cell_not_distinguished k0CounterResolution K0Counter

structure MaintenanceEvidence where
 obligationChecked : Bool
 supportAvailable : Bool
```

```
def nonVacuousMaintenance (m : MaintenanceEvidence) : Prop :=
 m.obligationChecked = true /\ m.supportAvailable = true

structure Realization where
 Carrier : Type
 admissible : Carrier -> Bool
 live : Carrier -> Bool
 cycle : Carrier -> Option CycleMode
 maintenance : Carrier -> Option MaintenanceEvidence

def maintenanceWitnessed (R : Realization) (x : R.Carrier) : Prop :=
 exists m, R.maintenance x = some m /\ nonVacuousMaintenance m

def cycleWitnessed (R : Realization) (x : R.Carrier) : Prop :=
 match R.cycle x with
 | none => False
 | some CycleMode.degenerate => maintenanceWitnessed R x
 | some _ => True

def supportWitnessed (R : Realization) (x : R.Carrier) : Prop :=
 cycleWitnessed R x \/ maintenanceWitnessed R x

def eligibleLive (R : Realization) (x : R.Carrier) : Prop :=
 R.admissible x = true /\ R.live x = true /\ supportWitnessed R x

theorem eligible_live_requires_cycle (R : Realization) (x : R.Carrier) :
 eligibleLive R x -> supportWitnessed R x := by
 intro h
 exact h.right.right

theorem eligible_live_requires_cycle_or_maintenance (R : Realization) (x : R.Carrier) :
 eligibleLive R x -> cycleWitnessed R x \/ maintenanceWitnessed R x := by
 intro h
 exact h.right.right

theorem eligible_live_requires_admissible (R : Realization) (x : R.Carrier) :
 eligibleLive R x -> R.admissible x = true := by
 intro h
 exact h.left

theorem cycle_mode_required_for_eligible_live (R : Realization) (x : R.Carrier) :
 eligibleLive R x -> (R.maintenance x = none) -> R.cycle x != none := by
 intro h hm
 cases h.right.right with
 | inl hc =>
 unfold cycleWitnessed at hc
 cases hcyc : R.cycle x with
 | none =>
 rw [hcyc] at hc
 cases hc
 | some c =>
 simp
 | inr hmnt =>
 rcases hmnt with  m, hsome, _
```

```
 rw [hm] at hsome
 cases hsome

structure Lifecycle (S Residue NewLive : Type) where
 admissible : S -> Bool
 live : S -> Bool
 death : S -> Bool
 deathBlocksLive : forall x : S, death x = true -> live x = false
 residueOf : S -> Option Residue
 rebirthOf : Residue -> Option NewLive
 identityInvariant : S -> NewLive -> Bool
 sourceToken : S -> Nat
 residueToken : Residue -> Nat
 newLiveToken : NewLive -> Nat
 sourceTokenInjective : forall x y : S, sourceToken x = sourceToken y -> x = y
 residueTokenInjective : forall x y : Residue, residueToken x = residueToken y -> x = y
 newLiveTokenInjective : forall x y : NewLive, newLiveToken x = newLiveToken y -> x = y

theorem declared_death_blocks_live {S Residue NewLive : Type}
 (L : Lifecycle S Residue NewLive) (x : S) :
 L.death x = true -> L.live x = false := by
 intro h
 exact L.deathBlocksLive x h

theorem death_live_conflict_impossible {S Residue NewLive : Type}
 (L : Lifecycle S Residue NewLive) (x : S) :
 L.death x = true -> L.live x = true -> False := by
 intro hdeath hlive
 have hblocked : L.live x = false := L.deathBlocksLive x hdeath
 rw [hlive] at hblocked
 cases hblocked

theorem residue_rebirth_are_typed_source_target_relations {S Residue NewLive : Type}
 (L : Lifecycle S Residue NewLive) (x : S) (r : Residue) (y : NewLive) :
 L.residueOf x = some r -> L.rebirthOf r = some y -> L.death x = true ->
 L.live x = false /\ L.residueOf x = some r /\ L.rebirthOf r = some y := by
 intro hres hreb hdeath
 exact And.intro (declared_death_blocks_live L x hdeath) (And.intro hres hreb)

theorem lifecycle_source_tokens_bind_carrier_endpoint {S Residue NewLive : Type}
 (L : Lifecycle S Residue NewLive) (x y : S) :
 L.sourceToken x = L.sourceToken y -> x = y := by
 exact L.sourceTokenInjective x y

theorem lifecycle_rebirth_tokens_bind_target_endpoint {S Residue NewLive : Type}
 (L : Lifecycle S Residue NewLive) (x y : NewLive) :
 L.newLiveToken x = L.newLiveToken y -> x = y := by
 exact L.newLiveTokenInjective x y

structure MorphismEvidence where
 sourceToken : Nat
 residueToken : Option Nat
 targetToken : Nat
 mclass : MorphismClass
 invariantPreserved : Bool
```

```
def isIdentityMorphism (m : MorphismEvidence) : Prop :=
 m.mclass = MorphismClass.identity /\
 m.invariantPreserved = true /\
 m.residueToken = none /\
 m.sourceToken = m.targetToken
```

1.8 Executable Finite-Model Evidence

The finite-model runner computes semantic verdicts from model facts and mutation controls. The public summary below omits internal publication-control rows and preserves the semantic proof evidence. **- verdict: - failure total:** 0 **- semantic evaluator:** True **- mutation control total:** 6 **- K-transition negative total:** 12

1.8.1 FM-T133-K0-RES-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.2 FM-T133-K0-RES-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.3 FM-T133-OMEGA-STATUS-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.4 FM-T133-OMEGA-STATUS-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.5 FM-T133-OMEGA-IDENTITY-EQUIVOCATION-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.6 FM-T133-K-ZERO-POS

- **case type:** theorem_case

- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.7 FM-T133-K-ZERO-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.8 FM-T133-K-ZERO-OBSTRUCTION-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.9 FM-T133-K-ZERO-LIVE-SUPPORT-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.10 FM-T133-BOUNDARY-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.11 FM-T133-BOUNDARY-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.12 FM-T133-HYBRID-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.13 FM-T133-HYBRID-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT

- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.14 FM-T133-HYBRID-NO-GUARD-STEP-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.15 FM-T133-HYBRID-NO-GUARD-STEP-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.16 FM-T133-HYBRID-GUARD-MISSING-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.17 FM-T133-HYBRID-GUARD-NONBOOLEAN-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.18 FM-T133-HYBRID-GUARD-VALUE-MISMATCH-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.19 FM-T133-HYBRID-RESET-SOURCE-MODE-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.20 FM-T133-HYBRID-RESET-TARGET-MODE-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT

- **failure total:**
- **witness:**

1.8.21 FM-T133-HYBRID-SMOOTH-CHART-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.22 FM-T133-HYBRID-SMOOTH-CHART-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.23 FM-T133-HYBRID-PROOF-UPDATE-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.24 FM-T133-HYBRID-PROOF-UPDATE-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.25 FM-T133-HYBRID-SMOOTH-LOCAL-LAW-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.26 FM-T133-HYBRID-PROOF-NO-RULE-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.27 FM-T133-DIM-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.28 FM-T133-DIM-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.29 FM-T133-CYCLE-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.30 FM-T133-CYCLE-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.31 FM-T133-ID-REBIRTH-NONIDENTITY-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.32 FM-T133-ID-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.33 FM-T133-ID-RESIDUE-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.34 FM-T133-ID-RESIDUE-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.35 FM-T133-ID-IDENTITY-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.36 FM-T133-MIN-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.37 FM-T133-MIN-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.38 FM-T133-KLEVEL-POS

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.39 FM-T133-KLEVEL-NEG

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.40 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.41 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.42 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.43 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.44 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.45 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.46 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.47 FM-T133-ID-TT-IDENTITY-INV-FALSE-END-TRUE-RES-TRUE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.48 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.49 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-FALSE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.50 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.51 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-FALSE-RES-TRUE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.52 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.53 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-FALSE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** ACCEPT
- **expected verdict:** ACCEPT
- **failure total:**
- **witness:**

1.8.54 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.55 FM-T133-ID-TT-IDENTITY-INV-TRUE-END-TRUE-RES-TRUE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.56 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.57 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-FALSE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.58 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.59 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-FALSE-RES-TRUE-CLAIM-TRUE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.8.60 FM-T133-ID-TT-RESIDUE-INV-FALSE-END-TRUE-RES-FALSE-CLAIM-FALSE

- **case type:** theorem_case
- **observed verdict:** REJECT
- **expected verdict:** REJECT
- **failure total:**
- **witness:**

1.9 Target-Blind Numeric Evidence

The empirical section reports bounded target-blind reconstruction rows over pinned official snapshots. These rows support model-core external review; they do not claim complete domain validation. - **lane total:** 5 - **failure total:** 0 - **support policy:** Rows are target-blind held-out reconstructions over pinned official snapshots. They support bounded numeric reconstruction claims only, not broad domain validation or novelty. - **domain validation policy:** NO_EMPIRICAL_PASS_FROM_OFFICIAL_SNAPSHOT_REPLAY_QA

1.9.1 Physics - OC133-TARGETBLIND-PHYSICS-001

- **snapshot:** validation/_raw/physics_nist_constants.txt
- **target-blind split:** Planck constant and speed of light rows are visible; inverse-meter joule relationship row is withheld until scoring
- **formula:** Planck_constant * speed_of_light
- **predicted value:** 1.9864458571489286e-25
- **observed value:** 1.986445857e-25
- **uncertainty:** 1e-33

- **residual:** 1.4892870576445277e-35
- **comparator baseline:** unit-incompatible Planck-constant-only negative control
- **comparator residual:** 1.9864458503739297e-25
- **negative control:** drop the speed-of-light factor and require a larger residual
- **falsifier:** Residual exceeds display-truncation tolerance or Planck-only control is not worse
- **snapshot hash:** 77fb90e66c40db3e6eb16630bc9c88e4c7c8beddbe5e71be406f2f26e3f67e67
- **replay hash:** a83ea273ef65d6136ad7c0dfe2f7d870822fe15a062a47ae906a7d46705a0cf1
- **support scope:** target-blind reconstruction of a held-out CODATA relationship from exact defining constants; not a novel physics law

1.9.2 Chemistry - OC133-TARGETBLIND-CHEMISTRY-001

- **snapshot:** validation/_raw/chemistry_pubchem_water.txt
- **target-blind split:** formula field is visible; MolecularWeight target is withheld until scoring
- **formula:** 2*atomic_weight(H)+atomic_weight(O)
- **predicted value:** 18.01528
- **observed value:** 18.015
- **uncertainty:** 0.02
- **residual:** 0.000280000000000058
- **comparator baseline:** CO2 molecular-weight negative control against water target
- **comparator residual:** 25.994500000000002
- **negative control:** replace H2O by CO2 and require larger residual
- **falsifier:** Residual exceeds declared uncertainty or CO2 control is not worse than formula reconstruction
- **snapshot hash:** 316ed3babcf1b5fd6580fc2922881bf553b5971de33b72f268d943529bce293f
- **replay hash:** c115a6d84fdad856976f60ad6d2fb4c9f9c397f994c7eafb09526f944004469d
- **support scope:** target-blind reconstruction of a held-out official snapshot field; not a novel chemistry law

1.9.3 Biology - OC133-TARGETBLIND-BIOLOGY-001

- **snapshot:** validation/_raw/biology_ncbi_geo_platform.txt
- **target-blind split:** NCBI ESearch retstart/idlist fields are visible; retmax pagination target is withheld until scoring
- **formula:** retstart + len(idlist)
- **predicted value:** 20.0
- **observed value:** 20.0
- **uncertainty:** 0.0
- **residual:** 0.0
- **comparator baseline:** use total hit count as pagination-size negative control
- **comparator residual:** 43988.0
- **negative control:** replace page-size reconstruction by total hit count and require a larger residual
- **falsifier:** Retmax differs from retstart plus returned id count or total-count control is not worse
- **snapshot hash:** f205645ec097af8972f7c70f4753a5d7eb154c4283e9caa454235843974bfda4
- **replay hash:** 6606a563b35116d6852bc936618757a29873c52e59541a0928546bc863cd4ecb
- **support scope:** target-blind reconstruction of a held-out NCBI/GEO API snapshot field; not a biological mechanism law

1.9.4 Systems - OC133-TARGETBLIND-SYSTEMS-001

- **snapshot:** validation/_raw/systems_world_bank_gdp.txt
- **target-blind split:** 2021-2023 train rows predict withheld 2024 World Bank WDI target
- **formula:** GDP_2023 + (GDP_2023-GDP_2021)/2
- **predicted value:** 111042234099108.1
- **observed value:** 110982661180013.0
- **uncertainty:** 1109826611800.1301
- **residual:** 59572919095.09375

- **comparator baseline:** last-observation carry-forward GDP_2023
- **comparator residual:** 4241013358949.0
- **negative control:** last-observation baseline must have larger residual
- **falsifier:** Held-out residual exceeds 1 percent of observed target or comparator is not worse
- **snapshot hash:** 52a3dcef732b262251662f20923d00ba8d4df239cce96097abf67ee6b15ff12b
- **replay hash:** 778ffef9592179e6062fad4b770d02612408603c70455f734c423475ff704be4
- **support scope:** retrospective target-blind holdout over pinned WDI rows; not a prospective macroeconomic law

1.9.5 Mathematics - OC133-TARGETBLIND-MATHEMATICS-001

- **snapshot:** proofs/FINITE_MODEL_CHECKS_1_3_3.json
- **target-blind split:** finite theorem-case rows are visible; aggregate machine_checked_subset_total is withheld until scoring
- **formula:** count_unique(theorem_id where case_type='theorem_case' and observed_verdict='ACCEPT' and passed=true)
- **predicted value:** 10.0
- **observed value:** 10.0
- **uncertainty:** 0.0
- **residual:** 0.0
- **comparator baseline:** positive_case_total negative control, which counts non-theorem support rows too
- **comparator residual:** 11.0
- **negative control:** replace theorem-id aggregate by positive_case_total and require a larger residual
- **falsifier:** Unique accepted theorem-case count differs from machine_checked_subset_total or broad positive-case control is not worse
- **snapshot hash:** 46ac711a2e86375ed333ce5c8a8b1b810f1f5b052fa657fb1508dbe7831b2f49
- **replay hash:** 050e21ffb0f2a5b73ea7faab51da781bbe66adc6bc6376941bb0a0e4eda293f2
- **support scope:** target-blind reconstruction of a finite proof-corpus aggregate; not a full-science program truth proof or empirical law

1.10 Journal Owner-Review Packages

Eight venue packets are included for owner review. They are not submitted by this release action. Each packet contains a package manifest, cover letter draft, checklist, reproducibility/data statement, conflict/funding statement, AI assistance disclosure, and venue-fit note. - **package total:** 8 - **recommended package total:** 2

1.10.1 FOUNDATIONS_OF_SCIENCE

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**

1.10.2 SYNTHESE

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**

1.10.3 FOUNDATIONS_OF_PHYSICS

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False

- **recommended:**

1.10.4 PHYSICAL_REVIEW_RESEARCH

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**

1.10.5 ACS_OMEGA

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**

1.10.6 ACTA_BIOTHEORETICA

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**

1.10.7 PLOS_COMPUTATIONAL_BIOLOGY

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**

1.10.8 GLOBAL_JOURNAL_OF_FLEXIBLE_SYSTEMS_MANAGEMENT

- **package status:** OWNER_REVIEW_READY_FOR_OWNER_REVIEW
- **submission allowed:** False
- **journal submissions allowed:** False
- **recommended:**